ABSTRACT

DEXTROUS represents the pinnacle of collaborative innovation in robotics, driven by a shared passion for technological advancement. Our project centers around the development of a sophisticated 6-degree-of-freedom (6-DOF) robotic arm, poised to redefine automation technology. The project team, comprising diverse experts in robotics and engineering, strategically selected the ESP32 and Raspberry Pi 3 platforms for their unparalleled versatility and adaptability to robotics applications.

At its core, DEXTROUS embodies a dual commitment to innovation and skill enhancement. Drawing upon our collective expertise in mechanical design, electronics, and programming, we navigate each challenge with unwavering determination and a spirit of resilience. Each obstacle encountered serves not as a roadblock but as a catalyst for personal and collective growth, pushing the boundaries of our technical prowess.

DEXTROUS symbolizes the fusion of passion, expertise, and ambition, creating a collaborative environment where exploration and discovery flourish. Through our combined efforts, we aim to transcend existing limitations and create a groundbreaking 6-DOF robotic arm that epitomizes the intersection of challenge and innovation.

In summary, DEXTROUS stands as a testament to our relentless pursuit of excellence in robotics. It is more than a project; it is a manifestation of our collective vision to revolutionize automation technology and shape the future of robotics through ingenuity, collaboration, and unwavering determination.

Vishal Sanjay Bhokare

Vedant Dinesh Butala

Sumedh Ravindra Kasat

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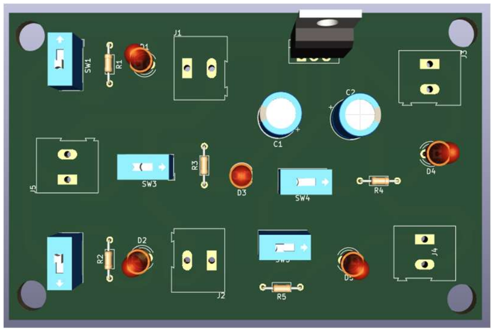
Feasibility report

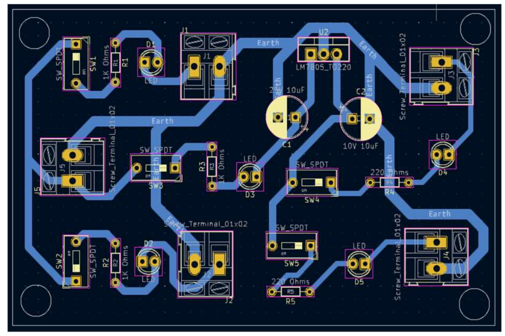
Title : DEXTROUS - 6 DOF Robotic Arm
Group members:

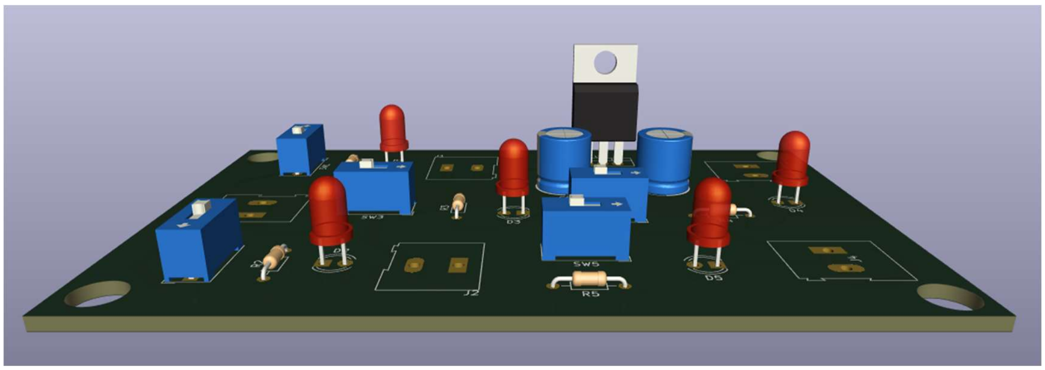
1) Vishal Sanjay Bhokare 2) Vedant Dinesh Butala 3)Sumedh Ravindra Kasat

Tools required	**Testing possibility**	**Controller**	**Cost**
Hardware/components	**Hardware** **Yes**	**ESP32** **PCA9685** **MG995/996** **LM2596S DC-DC Buck**	**Rs. 1648 /-**
Software	**Software** **Yes**	**1) Fusion 360** **2) KiCAD** **3) Arduino IDE**	**Free with Student License**
Tools available within campus or outside	**Sensors required**	**Signal conditioning if any**	**-**
1) Blue Wire Cutter **2) Electric Soldering Tool** **3) Breadboard** **4) Multimeter** **5) Wires and Jumper Cables** **6) Power Tools in Workshop** **7) CAD Lab for 3D Printing**	**The robotic arm employs key sensors for accurate operation: position encoders, proximity sensors, IMUs, current sensors, temperature sensors, and optional force sensors. Additionally, voltage/current and communication interface sensors enhance power management and data exchange efficiency.**	**Signal conditioning components enhance sensor performance by amplifying, filtering, converting, linearizing, calibrating, shifting voltage levels, isolating, and integrating signals for improved accuracy.**	**-**
Applications if any	**PCB design and fabrication**	**Datasheets/ application notes available**	**-**
The robotic arm finds applications in various fields such as manufacturing, automation, logistics, healthcare, and research. **Its versatility allows it to perform tasks like assembly, pick-and-place operations, material handling, surgical assistance, and experimentation in controlled environments.**	• **KiCad EDA 8.0.1** • **Flux.ai**	**Datasheets and application notes are essential references providing detailed specifications, characteristics, and usage guidelines for project components.** • **MG995/996** • **ESP32** • **PS4 Controller** • **PCA9685 Module**	**-**
Mechanical design	**Enclosure design**	**Demonstration**	**-**
The mechanical design encompasses the physical structure and components of the robotic arm, ensuring robustness, precision, and functionality.	**The enclosure design involves the creation of a protective casing or housing for the robotic arm components, safeguarding them from environmental factors and ensuring safe operation.**	**The demonstration involves highlighting the robotic arm's functionality, including its motion capabilities, interaction with external devices, and execution of predefined tasks.**	**-**

Title of project : DEXTROUS - 6 DOF Robotic Arm







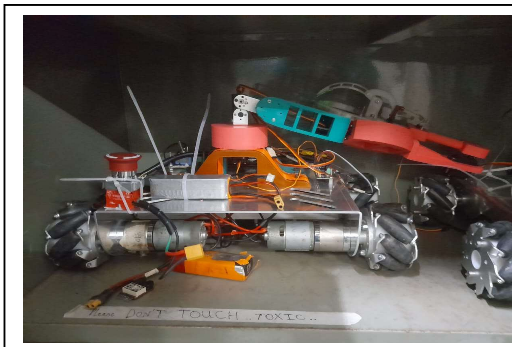
Electric specification

Input Specifications:
- Voltage Range: 12V DC
- Current Rating: 2A

Output Specifications:
- Voltage: 5V DC
- Current: Up to 1A

Features:
- Overcurrent & Short Circuit Protection
- Efficiency: >60%

Mechanical specification



High Efficiency, Compact Design, and Robust Protection Features.
Cost of System : Rs. 27,035
The cost of the system is competitively priced, offering exceptional value for its efficiency, reliability, and features.

Group members:

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32113	**Vedant Dinesh Butala**
32134	**Sumedh Ravindra Kasat**